

CASE STUDY 3: HEALTHCARE HYBRID CLOUD MIGRATION & HARDENING

Enterprise Healthcare Network Consolidation, Zero-Trust Identity, Storage Optimization & Deep Packet Network Analysis

PRIMARY DOMAIN Healthcare Cloud Architecture & Insider Threat Defense	NETWORK STACK Cisco Packet Tracer, Wireshark, IPsec IKEv2, Azure VPN
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I. PROJECT OVERVIEW & BUSINESS CHALLENGE

Client Profile & Infrastructure Silos: A multi-site healthcare network consisting of four physical eye care clinics operated as completely isolated infrastructure silos. Each facility maintained an independent on-premises database server hosting standalone CRM/EMR instances alongside localized directory shares holding massive diagnostic imaging portfolios (OCT scans and retinal photography).

The Catalyst (Insider Threat & Data Destruction): Prior to project engagement, a disgruntled former employee holding elevated domain administrator privileges intentionally corrupted a localized physical server at the flagship clinic. Because local NAS devices and external HDDs were continuously mounted to the corporate network with identical credentials, the actor wiped live production databases and backups simultaneously.

Operational Impact: Severe downtime forced reliance on manual paper tracking and partial cached software reconstruction, exposing fatal single points of failure, administrative privilege sprawl, unsegmented backup targets, a complete absence of air-gapped immutable storage, severely deteriorated 20-year-old brittle/cracking copper cabling, and hostile server housing (the flagship server was cramped inside a hot, dry HVAC room).

II. STAKEHOLDER DISCOVERY & INTERVIEW MATRIX

Stakeholder Role	Primary Concerns & Questions	Uncovered Pain Points	Design Impact & Resolution
Medical Director & Lead Optometrist	- How will traveling doctors access records? - Will scan loading times slow down care?	- Siloed patient records across 4 sites. - Inability to view historical OCT images.	Centralized 4 CRM databases into unified Azure VM topology backed by fast SSDs.
Director of Compliance & Risk	- How to guarantee backups can't be wiped? - Are we HIPAA compliant?	- Co-located, mounted NAS backups. - Shared local admin passwords.	Deployed Azure Recovery Services Vault with Immutable WORM locking & Soft Delete.
Practice Administrator & Bookkeeper (Doctor's Wife)	- Can we afford cloud hosting for multi-TB archives? - What is the capital outlay?	- High CapEx quote for physical servers. - Predictable cloud cost concerns.	Implemented Azure Blob Cool Tier, slashing storage costs ~60% under OpEx.
Clinical Network Administrator	- Why do large retinal files drop across VPNs? - Where are network bottlenecks?	Unstable S2S tunnels, severe MTU fragmentation, and brittle 20-year-old wiring in hot HVAC room.	Relocated server out of HVAC room, subcontracted rewiring, and optimized IPsec MTU.

III. FORMAL SCOPE BOUNDARY STATEMENT

IN-SCOPE Deliverables - Pre-design simulation in Cisco Packet Tracer and packet capture via Wireshark. - Azure VNet hub-and-spoke provisioning and Windows Server Datacenter VM hosting 4 DB instances. - Four active-active Site-to-Site IPsec (IKEv2) VPN tunnels with strict NSGs. - Azure Blob Storage (Cool Tier) integration for historical DICOM/OCT archives. - Server relocation out of the hot/dry HVAC room and subcontracted rewiring of all 20-year-old brittle copper lines. - Entra ID RBAC, MUA, Defender JIT, and Immutable Recovery Vaults.	OUT-OF-SCOPE Boundaries - Physical hardware replacement of localized optical cameras/diagnostic equipment. - Modifying third-party EMR application source code or schema architecture. - Provisioning consumer desktop hardware or workstation operating system upgrades. All workstations had to be replaced.
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■ IV. PRE-DESIGN NETWORK MAPPING & DIAGNOSTIC ANALYSIS

Utilized **Cisco Packet Tracer** to model inter-site routing and broadcast domain behavior across all four clinics, identifying routing loops and WAN bandwidth saturation. Deployed mirrored port monitoring (SPAN) with **Wireshark** and environmental auditing:

[Wireshark & Physical Diagnostic Capture Summary] 1. Retinal Image File Transfer Capture (50MB+ DICOM/OCT Scans): |— Frame Error Rate: 14.2% TCP Retransmissions across legacy WAN |— Path MTU Discovery: ICMP Fragmentation Needed (DF flag set) -> Packet Loss |— Effective Throughput: Throttled to 3.2 Mbps due to TCP Window Size exhaustion 2. Infrastructure & Environmental Audit: |— Server Housing: Core server cramped inside a hot, dry HVAC room causing thermal socket timeouts |— Cabling Condition: All enterprise wiring was 20 years old, brittle, cracking, and physically unstable

Resolution: Wireshark data and physical testing proved local network drops stemmed from TCP MSS/MTU mismatches, thermal stress in the HVAC room housing, and brittle 20-year-old wiring. MSS clamping (1350 bytes) was applied, IPsec tunnels were engineered, the server was relocated to a climate-controlled area, and a trusted local subcontractor replaced all deteriorated cabling.

■ V. CHRONOLOGICAL STEP-BY-STEP EXECUTION SEQUENCE

- P1 Audit & Baseline:** Mapped topology in Packet Tracer, captured PCAPs in Wireshark, and inspected HVAC room and 20-year-old brittle wiring.
- P2 Environmental & Rewiring Remediation:** Relocated flagship server out of the hot/dry HVAC room; subcontracted local vendor to replace all copper lines.
- P3 Azure VNet & VPN Provisioning:** Configured hub-and-spoke VNet and 4 Site-to-Site IPsec (IKEv2) tunnels with MSS clamping (1350 bytes).
- P4 VM & DB Consolidation:** Deployed Windows Server Datacenter VM hosting 4 isolated database instances for multi-tenant clinic separation.
- P5 Storage Tiering:** Configured Azure Managed SSDs for active DBs and Blob Storage (Cool Tier) for historical archives (-60% cost).
- P6 Zero-Trust Hardening:** Enforced Entra ID RBAC, MUA, and Defender for Cloud JIT Access (closing RDP/3389 ports).
- P7 Immutable Vault & Cutover:** Deployed Recovery Services Vault with WORM locking and Soft Delete; executed secure off-hours migration.

■ VI. KEY METRICS & RESUME HIGHLIGHTS

Metric Category	Pre-Migration Baseline	Post-Implementation Outcome
Server Footprint & Housing	4 Local servers (Flagship server in hot, dry HVAC room)	0 On-Premises Servers (100% Azure Cloud Migration)
Image Storage Cost	Expensive local NAS expansion (\$\$\$)	~60% Cost Reduction via Azure Blob Cool Tier
Backup Resilience	Mounted NAS/HDDs (Wiped by insider)	100% Immutable Vaults (MUA + Soft Delete)
WAN & Cabling Integrity	14.2% TCP Retransmissions / Brittle 20-yr copper	0% Retransmissions (Optimized IPsec & New Rewiring)

■ VII. RETROSPECTIVE REALITY, HIGH-PRESSURE PROCEEDINGS & OUTCOME

Capital Realities & Stakeholder Stress: Although financial outlay and operational disruption were significant—due to replacing workstations, relocating the server out of the hot/dry HVAC room, subcontracting brittle 20-year-old copper rewiring, and recovering fragmented legacy data—the lead stakeholder admitted a candid truth: he was simply trying to retire before ever having to face updating his ancient infrastructure.

High-Pressure Proceedings & Conflict Management: During intensive migration proceedings, the owner and lead stakeholder became aggressive and volatile under severe operational pressure. In response, our team calmly assured him that we were working methodically and deliberately to ensure absolute correctness so they would never be vulnerable to catastrophic data destruction again. While it is difficult to measure whether that reassurance fully defused the tension at the moment, the underlying architectural stability delivered permanent peace of mind.

Business Outcome & Ongoing Relationship: Clinicians gained instantaneous access to EMR portfolios and imaging scans without bottlenecks. Shortly following completion, the owner successfully leveraged the fully modernized, secure enterprise infrastructure to sell all four locations and the business for a substantial fee. Furthermore, despite the high-stress proceedings during the main migration, they have continued to retain our team for minor ongoing technical support and infrastructure maintenance issues, including installing printers and print servers.